

LIST OF PROBLEMS ABOUT BAYESIAN GAMES

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. Bach vs Stravinsky with incomplete information. Consider a variant of the BoS (Bach vs Stravinsky) I discussed in the Game Theory playlist. Suppose now that player 1 is unsure whether player 2 prefers to go out to the concert with her or prefers to avoid her, whereas player 2 knows player 1's preferences. Specifically, suppose that player 1 thinks that with probability $\frac{1}{2}$ player 2 wants to go out with her, and with probability $\frac{1}{2}$ player 2 wants to avoid her. The payoffs matrices are given below.

		Player 2	
		B	S
Player 1	B	2, 1	0, 0
	S	0, 0	1, 2

prob. $\frac{1}{2}$: 2 wishes to meet 1

		Player 2	
		B	S
Player 1	B	2, 0	0, 2
	S	0, 1	1, 0

prob. $\frac{1}{2}$: 2 wishes to avoid 1

- (a) Use a belief space to describe formally the belief configuration above.
 - (b) Compute all pure-strategy Bayes-Nash equilibria (BNE) from the game above.
 - (c) Are there any equilibria in behavioral (mixed) strategies? If so, compute all of them.
2. More BoS, more complications. Consider another variant of the BoS game, in which now, neither player knows whether the other wants to go out with her. Specifically, suppose that player 1 thinks that with probability $\frac{1}{2}$ player 2 wants to go out with her, and with probability $\frac{1}{2}$ player 2 wants to avoid her; player 2 thinks that with probability $\frac{2}{3}$ player 1 wants to go out with her and with probability $\frac{1}{3}$ player 1 wants to avoid her. This situation is partially represented by the matrices below. A state of the world is comprised of an ordered string of two characters where each character is either y or n ; each character is indicating whether the player in that position wants to go out with the other player by y , or wants to avoid the other player by n .

		Player 2	
		B	S
Player 1	B	2, 1	0, 0
	S	0, 0	1, 2

$\omega_1 = yy$

		Player 2	
		B	S
Player 1	B	2, 0	0, 2
	S	0, 1	1, 0

$\omega_2 = yn$

		Player 2	
		B	S
Player 1	B	0, 1	2, 0
	S	1, 0	0, 2

$\omega_3 = ny$

		Player 2	
		B	S
Player 1	B	0, 0	2, 2
	S	1, 1	0, 0

$\omega_4 = nn$

- (a) Use the four states of the world given to define type spaces and signalling functions so that your belief space models the situation described above. Notice that in the problem statement, I am already giving you the *posteriors* of each player, so your job will be to come up with the *common prior* that induces those posteriors.
 - (b) Compute *all* Bayes-Nash equilibria (pure and behavioral) for the game above.
3. Exchanging envelopes. Each of two individuals receives a sealed envelope. In each envelope, there is a ticket on which there is an integer number from 1 to m indicating the size of a prize she may receive. The players' tickets are assigned randomly and independently, and the probability of an individual's receiving each possible

number is positive. Each player can privately open the envelop, take a peek at the ticket, and then put it back in the envelope and seal it again. Then, each player is given the option to exchange the envelopes, and this option is given simultaneously to both players. Both individuals announce if they want to exchange the envelopes or not simultaneously. If both individuals wish to exchange the envelopes, then they are exchanged, otherwise each individual remains with their original envelope. Final prizes are awarded depending on the integer in the ticket inside the envelope under the possession of each individual.

- (a) Give, firstly, the description of the above situation as a Bayesian Game. Start by properly defining the belief space of the players and then add actions and payoffs to complete the formal description of the Bayesian Game.
 - (b) Show, that in any BNE of the game, the highest prize that either individual is willing to exchange is the smallest possible prize.
4. More infection! Consider the variant below of the infection game we analyzed in class. Type spaces are $T_1 = \{a, b, c\}$ and $T_2 = \{a', b'\}$. Signalling functions are given by $\tau_1(\omega_1) = a$, $\tau_1(\omega_2) = \tau_1(\omega_3) = b$, $\tau_1(\omega_4) = c$; $\tau_2(\omega_1) = \tau_2(\omega_2) = a'$, $\tau_2(\omega_3) = \tau_2(\omega_4) = b'$.

	L	R
U	2, 2	0, 0
D	3, 0	1, 1

ω_1 : prob. 27/40

	L	R
U	2, 2	0, 0
D	0, 0	1, 1

ω_2 : prob. 9/40

	L	R
U	2, 2	0, 0
D	0, 0	1, 1

ω_3 : prob. 3/40

	L	R
U	2, 2	0, 0
D	0, 0	1, 1

ω_4 : prob. 1/40

- (a) Answer the questions: What do the players know about which matrix they are at? What do they believe? You can start with an intuitive answer, but then use formal methods, i.e., analyze the situation using the knowledge operator, to formalize your intuition. How is this problem different from the one we analyzed in class?
 - (b) Show, using a method similar to the one we used in class (the *infection argument which is based on iterated dominance*) that the only BNE profile is when every type of player 1 chooses D and every type of player 2 chooses R .
5. Cournot with incomplete information about cost. Two firms compete in selling a good: one firm does not know the other firm's cost function. Assume that both firms can produce the good at a constant unit cost. Assume also that they both know that firm 1's unit cost is c , but only firm 2 knows its own unit cost; firm 1 believes that firm 2's cost is c_L with probability θ and c_H with probability $1 - \theta$, where $0 < \theta < 1$ and $c_L < c_H$. Assume also that the inverse demand function is $P(Q)$ with $P(Q) > 0$ if $Q \in [0, \bar{Q})$ and 0 otherwise. Assume also that $P(Q)$ is differentiable.
- (a) Model the above situation as a Bayesian Game. Give all the details.
 - (b) Characterize the set of pure strategy Bayes-Nash Equilibrium profiles. Characterize means that you just need to show me the *three equations* that identify an equilibrium profile, but you do not need to solve for them.
 - (c) Now assume that $P(Q) = \max\{\alpha - Q, 0\}$ for a parameter $\alpha > 0$. Notice that this satisfies the assumption on the inverse demand functions made in the general statement of the problem. Find the Bayes-Nash equilibrium profile for this game. Now you will actually need to solve the equations characterizing the equilibrium under the specific inverse demand I am giving you.
6. Cournot with incomplete information about both cost and information. Similar setup as the previous problem, but now suppose that firm 2 does not know whether firm 1 knows firm 2's cost. Specifically, suppose that firm 2 believes that one of the circumstances is that firm 1 knows its cost. I am going to help you model this situation: we need *four states of the world*:

$L0$: firm 2's cost is low and firm 1 does not know whether it is low or high.

$H0$: firm 2's cost is high and firm 1 does not know whether it is low or high.

$L1$: firm 2's cost is low and firm 1 knows it is low.

$H1$: firm 2's cost is high and firm 1 knows it is high.

Type spaces are $T_1 = \{0, L, H\}$ and $T_2 = \{L, H\}$; signalling functions are given by $\tau_1(L0) = \tau_1(H0) = 0$, $\tau_1(L1) = L$, $\tau_1(H1) = H$; $\tau_2(L0) = \tau_2(L1) = L$, $\tau_2(H0) = \tau_2(H1) = H$.

The posteriors are:

Firm 1: Type 0 assigns probability θ to state $L0$ and probability $1 - \theta$ to state $H0$; type L assigns probability 1 to state $L1$; type H assigns probability 1 to state H .

Firm 2: Type L assigns probability π to the state $L1$ and probability $1 - \pi$ to state $L0$; type H assigns probability π to state $H1$ and probability $1 - \pi$ to state $H0$.

- (a) Notice that I am giving only the posteriors when describing the beliefs of the players. Are these posteriors *consistent with a common prior*? That is, can the posteriors be induced through Bayes rule from a common prior after using the private information of the players to compute them? If so, what is the common prior?
- (b) Assume that $P(Q) = \max\{\alpha - Q, 0\}$ for a parameter $\alpha > 0$. Characterize the set of pure strategy Bayes-Nash equilibrium profiles. Again, this means that you need to show me the *six equations* that would identify an equilibrium profile, but you do not need to solve for them.

7. Free-riding and public good contribution. Consider the following public good production game. There are two players who may decide to produce or not a public good. The public good is produced if at least one player decides to contribute. Contribution is a 0-1 decision: either you contribute, or you don't, and the amount is fixed. The cost of contribution is privately known by the player (for example, the marginal utility of money, or the opportunity cost of contributing may be private knowledge), while the benefit of the public good is equal to 1 for everybody. Let $A_i = \{0, 1\}$ denote the action space of player i , where 0 means no contribution and 1 means contribution. Let c_i be the privately known cost of contribution, and assume that c_i is distributed uniformly on the interval $[0, 2]$ for each player and the c_i 's are independent for $i = 1, 2$. The payoff function for each player is:

$$u_i(a_i, a_j, c_i) = \begin{cases} 1 - c_i & \text{if } s_i = 1, s_j = 1 \\ 1 - c_i & \text{if } s_i = 1, s_j = 0 \\ 1 & \text{if } s_i = 0, s_j = 1 \\ 0 & \text{if } s_i = 0, s_j = 0 \end{cases}$$

- (a) Provide the belief space description of the situation above.
- (b) Suppose that $c_i > 1$. Does any player have an interim-dominated action?
- (c) Assume that $c_i \leq 1$. Let q_j be the probability that player j contributes. What is the expected utility of contributing for player i ? What is the expected utility of not contributing for player i ?
- (d) Using the previous item, when is it strictly optimal for player i to contribute? When it is strictly optimal for player i to not contribute? Note that you should get some conditions involving c_i and q_j in your answer.
- (e) Using the previous item, argue that, since a symmetric reasoning applies to both players, all Bayes-Nash equilibria must be characterized by two thresholds (c_1^*, c_2^*) such that types $c_i < c_i^*$ contribute while types $c_i > c_i^*$ do not contribute.
- (f) Using the above, compute q_j^* , which is the equilibrium probability that player j contributes.
- (g) Finalize characterizing the equilibrium using the following: notice that the two thresholds (c_1^*, c_2^*) are such that, *at those exact types, the players are indifferent* between contributing and not contributing.
- (h) What is the equilibrium probability of freeriding, i.e., at the equilibrium strategies, what is the probability that both players find the public good beneficial, the public good is produced but only one of the players is contributing?